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Department of Information Technology

A.Y. 2025-2026
Class: BE-IT A/B, Semester: VIII
Subject: Cloud Computing Lab

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Experiment – 9: Implement Containerization

Aim: To study and Implement Containerization using Docker on AWS/Azure/Google cloud platform.(Free Tier Cloud Platforms).

Objective: After performing the experiment, the students will be able to learn –
- How to implement containerization using docker

Lab objective mapped : ITDO8024.5 : To comprehend Docker and its role in the DevOps lifecycle.

Prerequisite: An AWS account

Requirements: Cloud Login**,** Desktop, Browser, Internet

Pre-Experiment Theory: (describe following topics in soft copy form)
1. **Amazon Elastic Container Service (Amazon ECS)**
2. **Docker applications**
3. **Amazon ECS task definitions**
4. **Application Load Balancer**

Procedure:

Extended Theory: (to be written in hand)
1. How is Amazon ECS different from AWS Elastic Beanstalk?

Results/Calculations/Observations:
Fill the following observation tables-(to be written in hand)

Amazon Aurora Vs RDS

Sr. No	**Questions**	**Yes/No**	**Justify**
1.	Does Amazon ECS support any other container types?		
2.	Does Amazon ECS support batch jobs?		
3.	Can user use their own Amazon Machine Image (AMI) on ECS?		

Conclusion:
1. Write what was performed in the experiment
2. Mention a few applications of what was studied.
3. Write the significance of the studied topic

References:
[1] [Online]https://ahireharshal.com/cloud/aws-ecs/
[2] [Online]https://docs.docker.com/cloud/ecs-integration/

Pre-Experiment Theory:

1. Amazon Elastic Container Service (Amazon ECS)

1. Amazon Elastic Container Service is a fully managed container orchestration service provided by Amazon Web Services.
2. It is used to run, stop, and manage Docker containers on a cluster of virtual machines.
3. Amazon ECS eliminates the need to install and operate container orchestration software manually.
4. It supports both serverless computing using AWS Fargate and EC2-based infrastructure.
5. ECS allows users to define applications, allocate resources, and manage scaling automatically.
6. It integrates easily with other AWS services such as load balancers, IAM, and monitoring tools.
7. It is widely used for microservices architecture and scalable application deployment.

2. Docker applications

1. These containers ensure that the application runs consistently across different environments.
2. Docker applications include the application code, runtime, libraries, and system tools required for execution.
3. Containers are lightweight and faster compared to traditional virtual machines.
4. Docker applications can be easily deployed, scaled, and moved between development, testing, and production environments.

5. They support microservices architecture by allowing multiple small services to run independently.
6. Docker improves portability, efficiency, and resource utilization in modern application development.

3. Amazon ECS task definitions

1. An Amazon ECS Task Definitions is a blueprint that describes how a Docker container should run in Amazon ECS.
2. It specifies details such as container image, CPU and memory requirements, and networking configuration.
3. Task definitions define one or more containers that form a task in ECS.
4. They include information about port mappings, environment variables, and storage volumes.
5. Each task definition is versioned, allowing updates and rollback if needed.
6. Tasks are launched based on these definitions, ensuring consistency across deployments.
7. Task definitions play a crucial role in managing and scaling containerized applications.

4. Application Load Balancer

1. The Application Load Balancer is a service provided by AWS to distribute incoming application traffic across multiple targets.
2. It operates at the application layer (Layer 7) of the OSI model.
3. The load balancer routes traffic based on content such as URL paths, hostnames, or HTTP headers.
4. It improves application availability and fault tolerance by distributing requests across multiple containers or servers.
5. It supports dynamic port mapping, which is useful for container-based applications running on ECS.
6. The Application Load Balancer integrates seamlessly with Amazon ECS to route traffic to running tasks.
7. It also provides features like SSL termination, health checks, and automatic scaling.